

1. The "Smart Tool" Router

In Agentic AI, the system must decide which "tool" (function) to call based on the user's input.

- **The Task:** Create two functions: `get_weather()` and `get_time()`. Write a main loop that asks for user input. Use conditions to call the correct function if the user mentions "weather" or "time."
- **Sample Input:** "What is the weather like?"
- **Sample Output:** "Calling Weather Tool..."

2. The Auto-Retry Wrapper

Sometimes an AI tool fails (like a bad API connection). You need a function that tries to run a task multiple times before giving up.

- **The Task:** Write a function `attempt_task()`. Inside, use a loop that tries to complete a task 3 times. If it fails all 3 times (use `input()` to simulate success/failure), return "Task Failed."
- **Sample Input:** User types "fail", "fail", "success".
- **Sample Output:** "Task completed on attempt 3."

3. The Prompt Sanitizer Function

Description: Before an AI processes a list of prompts, they need to be cleaned (spaces removed, converted to lowercase).

- **The Task:** Write a function `clean_text(text)`. Then, take a list of messy strings and use a loop to pass each one through the function and store the clean versions in a new list.
- **Sample Data:** [" HELLO ", "pyThOn ", " AI "]
- **Sample Output:** ["hello", "python", "ai"]

4. The Token Cost Validator

Agents need to check if a task is "too expensive" before running it.

- **The Task:** Write a function `calculate_cost(word_count)`. It should return `word_count * 0.001`. In your main code, loop through a list of different paragraph lengths. If the cost is $> \$0.05$, use a condition to "Reject" it.

- **Sample Data:** [10, 100, 500]
- **Sample Output:** Cost 0.01: OK, Cost 0.1: Too expensive!

5. The Multi-Agent "Consensus"

Sometimes you ask three different "AI Agents" (functions) for an answer and take the majority vote.

- **The Task:** Create three functions that each return either True or False. Use a loop to collect their results into a list. Use a condition to check if the count of True is 2 or more.
- **Sample Output:** "Majority says: Proceed."